IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

SPEECH PROCESSING

Friday, 15 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : W. Dai

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Speech Processing

1. A speech signal $s(n)$ is known to have been corrupted by additive noise $\nu(n)$. The noisy speech signal can be written in the time domain as $y(n) = s(n) + \nu(n)$ and in the frequency domain as $Y(l,k) = S(l,k) + V(l,k)$ for time-frame index l and frequency index k.

   a) i) Given a conceptual explanation of what is meant by the spectral variance of a signal in this context. [3]

   ii) Give the mathematical definition for the spectral variance of $y(n)$, denoted $\phi_y(l,k)$. [3]

   b) In a particular speech enhancement application, it is desired to estimate the noise spectral variance. Now, assume that it is only possible to observe $y(n)$.

   i) One method to compute an estimate of the spectral variance of the noise, $\hat{\phi}_v(l,k)$, employs a Voice Activity Detector (VAD). Describe and explain this technique in detail. [2]

   ii) State one other method to compute an estimate of the spectral variance of the noise. Describe and explain this technique in detail and give mathematical expressions for the estimators. [2]

   iii) Compare and contrast the advantages and disadvantages of these alternative methods. [3]

   c) Consider a device which performs noise reduction on a noisy speech signal. The device uses the method of amplitude spectral subtraction.

   i) Show that the output signal of the device at time-frame l and frequency index k can be written

$$Z(l,k) = H(l,k)Y(l,k)$$

   and write down an expression for $H(l,k)$ in terms of $\hat{\phi}_v(l,k)$. [3]

   ii) It is not known what method the device uses to form the estimate $\hat{\phi}_v(l,k)$. Design a test signal to probe the device such that, by observing the output of the device in response to your test signal input, the method of estimation of $\hat{\phi}_v(l,k)$ can be deduced. [4]

2. a) Describe and explain 3 measures for the objective (sometimes called 'instrumental') assessment of the quality of speech in a speech coding application. In your answer, include any relevant mathematical definitions as well as an explanation of each measure. [6]

b) The block diagram of the main elements of a speech coder is shown in Fig. 2.1. Consider a general speech signal $s(n)$ to be encoded.

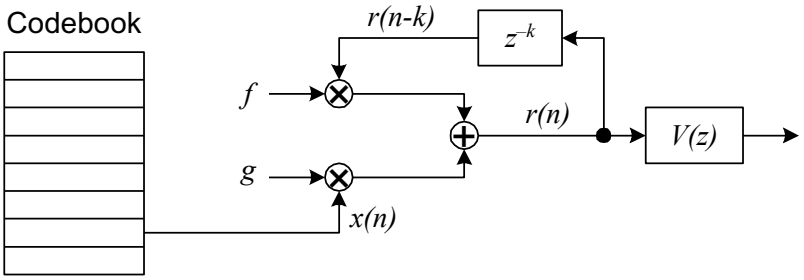


Figure 2.1

i) By making specific reference to Fig. 2.1, identify and explain the operation of the parts of this coder that are responsible for modelling the periodic components and the non-periodic components of the excitation signal $r(n)$. [2]

ii) This coder is designed to determine the values of k and f so as to minimize the energy function

$$E(k,f) = \sum_{n=0}^{N-1} (r(n) - r(n-k)f)^2 .$$

Assume that the value of k is given and ˆ indicates the optimal value. Show that the optimum values $\hat{f}$ and the corresponding energy function $\hat{E}$ are given by

$$\hat{f}(k) = \frac{\sum_{n=0}^{N-1} r(n)r(n-k)}{\sum_{n=0}^{N-1} r^2(n-k)}$$

and

$$\hat{E}(k) = \sum_{n=0}^{N-1} r^2(n) - \frac{\left(\sum_{n=0}^{N-1} r(n)r(n-k)\right)^2}{\sum_{n=0}^{N-1} r^2(n-k)}.$$

[6]

iii) In a particular operation of this speech coder, the signal $r(n)$ has the following values and is zero for all other n.

$$\begin{aligned} r(-100) &= 9 \\ r(-95) &= 4 \\ r(-50) &= 10 \\ r(-45) &= 7 \\ r(0) &= 5 \\ r(5) &= 8 \end{aligned}$$

For $N = 40$, determine $\hat{f}(k)$ and the corresponding values of $\hat{E}(k)$ for $k = 45$ and $k = 50$. [6]

3. a) Describe Linear Predictive Coding (LPC) and its use in speech signal processing applications. Your description should summarise the main concepts in text and, where relevant, diagrams. It is not necessary to give or derive any mathematical formulae. [6]

b) A linear predictor for the speech signal $s(n)$ can be written

$$s(n) = Gu'(n) + \sum_{j=1}^{p} a_j s(n-j).$$

Explain what the terms $u'(n)$, G and a_j represent in this expression. [3]

c) Formulate a time domain expression for the prediction error and hence show that the prediction error is minimum when the following expression is satisfied:

$$\sum_{j-1}^{p} a_j \sum_{n\in\{F\}} s(n-j)s(n-i) = \sum_{n\in\{F\}} s(n)s(n-i) \quad \text{for} \quad i = 1,\ \ldots,\ p$$

where $\{F\}$ denotes the set of speech samples within a frame. [4]

d) i) Using mathematical notation and conceptual description, define the complex cepstral coefficients of a speech signal and briefly state their important properties for speech processing. [2]

ii) Derive a recursion to obtain the complex cepstral coefficients from the terms a_j as defined by parts b) and c) of this question. [4]

iii) Find the complex cepstral coefficients for the case when

$$[a_0,\ a_1,\ a_2,\ a_3,\ a_4] = [1.00,\ -0.20,\ -0.25,\ 0.18,\ 0.40].$$

[1]

4. a) i) Draw a block diagram showing the main elements of the

Listen-Attend-Spell (LAS)

neural network architecture in a form that could be used in automatic speech recognition. Label the block diagram and explain the meaning of the labels. [3]

ii) The mel frequency scale is used in the process of computing mel-frequency cepstral coefficients (MFCCs). These MFCCs may be used as the input speech features for automatic speech recognition systems. The mel scale is defined as

$$\text{mel}(f) = 2595 \log_{10}\left(1 + \frac{f}{700}\right)$$

where f is the frequency measured in Hz.

Sketch a graph of the mel scale against frequency for the range $0 \leq f \leq 8000$ Hz. [2]

Explain why the MFCCs are usually a good choice of speech features for automatic speech recognition. [2]

b) In a speech recognition system, an observation sequence comprising T frames $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T\}$ is to be compared with a hidden Markov model (HMM) containing S states. For the hidden Markov model, the probability density of generating an observation frame $\mathbf{x}$ from state s is $d_s(\mathbf{x})$ and the transition probability from state i to state j is a_{ij}.

i) Now let $B(t,s)$ be defined as the best (maximum) probability density of generating the sequence $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T\}$ from any sequence of states for which frame 1 is aligned in state 1 and frame t is in state s. Explain how, for $t > 1$, $B(t,s)$ can be expressed in terms of $B(t-1,i)$ for $i = 1, 2, \ldots, S$. Indicate the values that should be given to $B(1,i)$. [6]

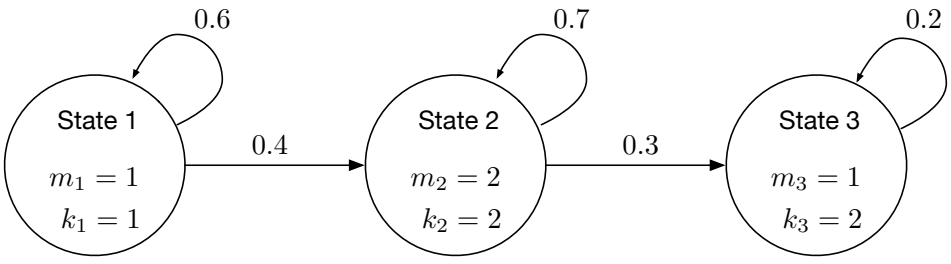


Figure 4.1

ii) Figure 4.1 shows the transition probabilities for a 3-state hidden Markov model. The input feature vector contains only a single element, and its probability density in frame s is given by:

$$d_s(x) = \frac{\exp\left(-|x - m_s|/k_s\right)}{2k_s}$$

where m_s and k_s are state-dependent parameters given in the figure.

Consider the alignment of the observation sequence $\{x_1, x_2, \ldots, x_5\} = \{1, 1.5, 2, 1, 1\}$ and hence determine the value of $B(5,3)$ and the state sequence to which it corresponds. You should perform all your calculations to 5 decimal places. [7]